THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHEMATICS TWO-CHAPTER TEST SET C

Chapters 3 & 4: **Pair of Linear Equations in Two Variables, Quadratic Equations** | RBSE Board Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Use graph paper wherever a graph is asked.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The graph of $y = -3$ is a line: 1

(a) through the origin **(b)** the x-axis

(c) parallel to the x-axis **(d)** parallel to the y-axis

2. If $x = 1,\ y = 1$ is a solution of $9ax + 12ay = 63$, then a is: 1

(a) 7 **(b)** 1 **(c)** 21 **(d)** 3

3. The pair $5x - 8y + 1 = 0$ and $3x - \frac{24}{5}y + \frac{3}{5} = 0$ has: 1

(a) two solutions **(b)** a unique solution

(c) no solution **(d)** infinitely many solutions

4. A student states: "If $\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$, the lines are parallel." The correct response is: 1

(a) They coincide **(b)** Correct

(c) Incorrect — the lines intersect **(d)** Cannot be determined

5. For $x + y = 8$ and $x - y = 2$, Student A gets $(5, 3)$ and Student B gets $(3, 5)$. Then: 1

(a) Student B is correct **(b)** Student A is correct

(c) Both are correct **(d)** Both are wrong

6. A root of $2x^2 + x - 6 = 0$ is: 1

(a) 3 **(b)** $-\frac{3}{2}$ **(c)** 2 **(d)** $\frac{3}{2}$

7. The discriminant of $\sqrt{3}x^2 + 2\sqrt{2}x - 2\sqrt{3} = 0$ is: 1

(a) −16 **(b)** 16 **(c)** 8 **(d)** 32

8. The roots of $4x^2 - 12x + 9 = 0$ are: 1

(a) real and distinct **(b)** real and equal

(c) zero **(d)** not real

9. A student states: "$x^2 + 3x + 5 = 0$ has real roots because all its coefficients are positive." The correct response is: 1

(a) It has one root **(b)** Correct

(c) Incorrect — $D = 9 - 20 < 0$ **(d)** Cannot be determined

10. "The product of two consecutive odd integers is 63" is represented by: 1

(a) $x^2 + 2x - 63 = 0$ **(b)** $x^2 - 63 = 0$ **(c)** $x^2 + x - 63 = 0$ **(d)** $2x + 2 = 63$

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Solve: $\frac{x}{2} + \frac{2y}{3} = -1$ and $x - \frac{y}{3} = 3$. 2

12. Given $2x + 3y - 8 = 0$, write another linear equation so that the pair forms 2
(i) parallel lines
(ii) coincident lines.

13. Solve by factorisation: $6x^2 - x - 2 = 0$. 2

14. Find k if $x^2 - 2(k+1)x + k^2 = 0$ has equal roots. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Six years hence a man's age will be three times his son's age, and three years ago he was nine times as old as his son. Find their present ages. 3

16. The sum of the squares of two consecutive natural numbers is 421. Find the numbers. 3

17. Draw the graphs of $y = x$ and $y = -x + 4$. Find the vertices and area of the triangle formed by these lines and the x-axis. 3

OR

2 tables and 3 chairs cost ₹3500, and 3 tables and 2 chairs cost ₹4000. Find the cost of a table and of a chair. 3

18. Solve for x: $4x^2 + 4bx - (a^2 - b^2) = 0$. 3

OR

Find two numbers whose sum is 15 and the sum of whose reciprocals is $\frac{3}{10}$. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study:** An electrician charges a fixed visit fee plus an hourly rate. A 3-hour job costs ₹550 and a 5-hour job costs ₹850. 5
(i) Form the equations.
(ii) Find the visit fee and hourly rate.
(iii) Find the charge for 8 hours.
(iv) Can a bill be ₹700 for a whole number of hours?

OR

Solve graphically: $2x + y = 8$ and $x - y = 1$. Also find the points where the lines meet the y-axis and the area of the triangle formed with the y-axis. 5

20. **Case study:** For a function, 300 chairs are arranged in rows. The number of chairs in each row is 5 more than the number of rows. 5
(i) Form the equation.
(ii) Find the number of rows and chairs per row.
(iii) Is it possible to arrange 350 chairs in the same way? Justify.

OR

A car covers 300 km at uniform speed. If the speed is increased by 10 km/h, it takes 1 hour less. Find the original speed and verify. 5

END OF QUESTION PAPER — Total: 40 Marks